



Module 2 challenge

1st Attempt:

1. In Python, when does an `else` statement execute a piece of code?

1 / 1 point

- ☐ When the `if` statement contains numeric data
- ☐ When the `if` statement contains a true condition
- ☒ When the `if` statement contains a false condition
- ☐ When the `if` statement contains text data

✓ Correct

2. In Python, what is the process of restructuring code while maintaining its original functionality?

1 / 1 point

- ☒ Refactoring
- ☐ Reprogramming
- ☐ Branching
- ☐ Converting

✓ Correct

3. A data professional wants to define a function to calculate the area of a rectangle. What code should they begin with?

1 / 1 point

- ☐ `if area_rectangle(length, width):`
- ☐ `else area_rectangle(length, width):`
- ☒ `def area_rectangle(length, width):`
- ☐ `return area_rectangle(length, width):`

✓ Correct

4. Which Python keyword makes a function produce new results and save the results for later use?

1 / 1 point

- ☐ as
- ☐ def
- ☒ return
- ☐ if

✔ Correct

5. Fill in the blank: To reduce errors and enhance teamwork, data professionals employ clean code best practices such as _____ and modularity.

1 / 1 point

- ☐ redundancy
- ☒ reusability
- ☐ immutability
- ☐ branching

✔ Correct

6. What is the Python comparator for *not equal to*?

1 / 1 point

- ☐ ==
- ☐ >=
- ☐ <=
- ☒ !=

✔ Correct

7. Fill in the blank: In Python, the `if` statement branches the execution based on a specific condition being _____.

1 point

- ☐ true
- ☒ false
- ☐ equal
- ☐ not equal

✘ Incorrect

Review [the video about branching.](#) [🔗](#)

Answer: True

8. Which code block(s) will yield results that are identical to those of the following function? Select all that apply.

0.8 / 1 point

```
1 def is_even(x):
2     if x % 2 == 0:
3         return True
4     else:
5         return False
```



```
1 def is_even(x):
2     if x % 2 != 0:
3         return False
4     else:
5         return True
```



```
1 def is_even(x):
2     if x % 2 == 0:
3         return True
4     return False
```



```
1 def is_even(x):
2     if x % 2 != 0:
3         return True
4     return False
```



```
1 def is_even(x):
2     return x % 2 == 0
```

✔ Correct

9. A data professional writes the following code: `print(not 4 == 'Data')`. What result will Python display when they run the code?

1 / 1 point

- ☒ True
- ☐ False
- ☐ Equal
- ☐ Not equal

✓ Correct

10. Fill in the blank: A data professional can add a _____ to the beginning of a function's body to summarize the function's behavior and explain its arguments and return values.

1 / 1 point

- ☒ docstring
- ☐ comparator
- ☐ conditional statement
- ☐ logical operator

✓ Correct

2nd Attempt:

1. Fill in the blank: In Python, the _____ statement sets a piece of code to run only when the condition of the `if` statement is false.

1 / 1 point

- ☐ `then`
- ☒ `else`
- ☐ `if`
- ☐ `return`

✓ Correct

2. What is the term for code that is written to be readable and to make its purpose clear?

1 / 1 point

- ☒ Self-documenting
- ☐ Open source
- ☐ Branching
- ☐ Immutable

✓ Correct

3. A data professional wants to define a function to calculate the square of a number. What code should they begin with?

1 / 1 point

- ☒ `def find_square(n):`
- ☐ `if find_square(n):`
- ☐ `else find_square(n):`
- ☐ `return find_square(n):`

✓ Correct

4. Which Python keyword makes a function produce new results and save the results for later use?

1 / 1 point

- ☐ `if`
- ☐ `as`
- ☐ `def`
- ☒ `return`

✓ Correct

5. What are best practices for writing clean code? Select all that apply.

0.8 / 1 point

☒ Modularity

✓ Correct

☐ Clarity

☐ Redundancy

☒ Reusability

✓ Correct

✗ You didn't select all the correct answers

Answer: Clarity (also)

6. What is the Python comparator for *less than or equal to*?

1 / 1 point

☐ >=

☐ ==

☒ <=

☐ !=

✓ Correct

7. Fill in the blank: In Python, the _____ statement branches the execution based on a specific condition being true.

1 / 1 point

☐ elif

☒ if

☐ then

☐ else

✓ Correct

8. Which code block will yield results identical to those of the following function?

1 / 1 point

```
1 def is_odd(x):  
2     if x % 2 == 0:  
3         return False  
4     else:  
5         return True
```

☐

```
1 def is_odd(x):  
2     return True  
3     if x % 2 == 0:  
4         return False
```

☐

```
1 def is_odd(x):  
2     return True  
3     if x % 2 == 0:  
4         return False
```

☐

```
1 def is_odd(x):  
2     return False if x % 2 == 0  
3     else:  
4         return True
```

☒

```
1 def is_odd(x):  
2     if x % 2 == 0:  
3         return False  
4     return True
```

✔ Correct

Answer: Option B is also correct

9. A data professional writes the following code: `print(33 > 12 or 9 < 7)`. What result will Python display when they run the code?

1 / 1 point

- ☒ True
- ☐ Not equal
- ☐ False
- ☐ Equal

✓ Correct

10. A data professional adds a docstring to the beginning of a function's body. What is the purpose of the docstring?

1 / 1 point

- ☐ To define the function
- ☒ To summarize the function's behavior and explain its arguments and return values
- ☐ To run the function and print the results
- ☐ To make the function produce new results and save the results for later use

✓ Correct